

Reliable API Error Management Workflow

Lesson 5 Assessment — Documentation Package

Project: Reliable API Workflow Blueprint

This document describes the implemented n8n workflow for retry protection, inline API error handling, status-code routing, centralized error monitoring, and incident reporting.

1. Workflow Architecture

Main workflow: Manual Trigger → HTTP_RetryOnFailure → HTTP_TrapErrorsInline → IF_IsError404 → Log_ProfileNotFound / Log_Success.

Centralized monitoring workflow: Error Trigger → Build_Incident_Report → Log_SlackAlert.

2. Node Configuration

Node	Purpose / Configuration
Manual Trigger	Starts the reliability workflow manually.
HTTP_RetryOnFailure	HTTP request with Retry on Failure enabled; maximum 3 tries; 2-second retry interval; timeout configured to 10 seconds.
HTTP_TrapErrorsInline	HTTP request configured with full response, never-error behavior, and continue-on-error handling so downstream routing is not blocked.
IF_IsError404	Checks statusCode and routes HTTP 404 responses to the profile-not-found path; other responses continue to the success path.
Log_ProfileNotFound	Creates a structured handled record containing handled status, reason, status code, error body, and timestamp.
Log_Success	Creates a structured result for successful or non-404 responses with handled status, reason, status code, and timestamp.
Error Trigger	Centralized workflow trigger for unhandled workflow failures.
Build_Incident_Report	Builds failedNode, executionId, workflowName, errorMessage, and incidentTime.
Log_SlackAlert	Sends the structured incident information as an operational Slack alert.

3. Error Handling Design

The workflow separates handled API responses from unhandled workflow failures. The inline HTTP node preserves the response for downstream inspection, while the IF node specifically identifies HTTP 404 responses. A 404 is logged as a handled profile-not-found condition. Other responses are routed to the success/non-404 logging path.

The centralized Error Trigger workflow is intended for failures that escape normal inline handling. It converts the error event into an operational incident report and forwards the report to the configured Slack alert step.

4. Retry, Timeout and Recovery Strategy

The primary HTTP request is configured with retry-on-failure behavior, a maximum of three attempts, and a two-second retry interval. The HTTP request nodes also have a 10-second timeout so slow requests do not block execution indefinitely.

Assessment verification note: the exported workflow clearly records the three-attempt retry configuration and 2-second interval. Exponential-backoff behavior should be verified in the n8n node UI before final submission if the assessor requires the setting to be explicitly enabled rather than inferred from retry configuration.

5. Incident Reporting Design

Field	Purpose
failedNode	Identifies the last executed node associated with the failure.
executionId	Identifies the affected n8n execution.
workflowName	Identifies the workflow where the incident occurred.
errorMessage	Captures the available error message.
incidentTime	Records the incident timestamp.

The alert step formats these fields into a workflow failure notification for the configured Slack channel.

6. Test Cases

Test	Expected Behavior
HTTP 200 / successful response	IF_IsError404 evaluates false and Log_Success records the response.
HTTP 404	IF_IsError404 evaluates true and Log_ProfileNotFound records the handled 404.
HTTP 429	Retry-capable request behavior should be observed and documented during execution testing.
HTTP 503	Transient failure should be evaluated against the retry configuration.
Network interruption	Retry/timeout behavior should prevent indefinite execution.
Authentication failure	Failure should be reviewed as a non-retryable or configuration issue and monitored through the appropriate path.
Slow API response	10-second timeout should terminate an excessively slow request safely.
Unhandled workflow error	Error Trigger should create an incident report and send the configured alert.

7. Governance and Documentation

Descriptive node names are used throughout the workflow to make the reliability behavior clear during review. The architecture separates inline API response handling from centralized monitoring, supporting easier troubleshooting and operational ownership.

8. Execution Evidence

A successful execution was captured for the main workflow, showing a 200 status response reaching Log_Success. The execution screenshot can be included in the GitHub repository and LMS evidence package. A separate 404 execution should be captured for explicit ProfileNotFound routing evidence if not already available.

9. Known Limitations / Assumptions

The exported workflow contains placeholder API URLs that must be replaced with valid test endpoints before final reproducibility testing. The Slack alert node also requires a configured Slack credential and channel. The documentation does not claim successful validation of scenarios that were not explicitly executed.

10. Submission Package

- Exported n8n workflow JSON
- Centralized Error Monitor workflow JSON
- Execution screenshots / test evidence
- This PDF documentation package
- Public GitHub repository containing source and documentation
- Loom/YouTube demonstration link
- LMS description with challenges, resolutions, assumptions, limitations, and improvements

Source basis: This documentation was prepared from the uploaded Reliable API Workflow and Centralized Error Monitor n8n workflow exports.